2026 INTERNATIONAL NEAR EAST CONFERENCE
ON ELECTRICAL & ENERGY SYSTEMS
21–23 October 2026 · Dubai, United Arab Emirates · Undergraduate Innovation Challenge

SMART CITIES · SUSTAINABLE ENGINEERING

Encirra

A coupled power + water digital twin for verified recovery from cascading infrastructure failures

One fault. Connected consequences. Verified recovery decisions.

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A grid-safe recovery is not a water-safe recovery.

A Feeder F3 fault stops the pump that supplies the hospital's zone. Its tank hides the problem for 72 minutes. Conventional restoration verifies electrically and still lets the hospital lose water; Encirra's plan keeps it supplied.

HOSPITAL WITHOUT WATER
79 → 0 min
Grid-only → Encirra · feeder-fault scenario, 3 h horizon

WATER DEMAND SERVED
78.4 → 100 %

CRITICAL FACILITIES RECOVERED
0/3 → 3/3

LOAD RESTORED · HARD VIOLATIONS
69.2 → 78.7 % · 0

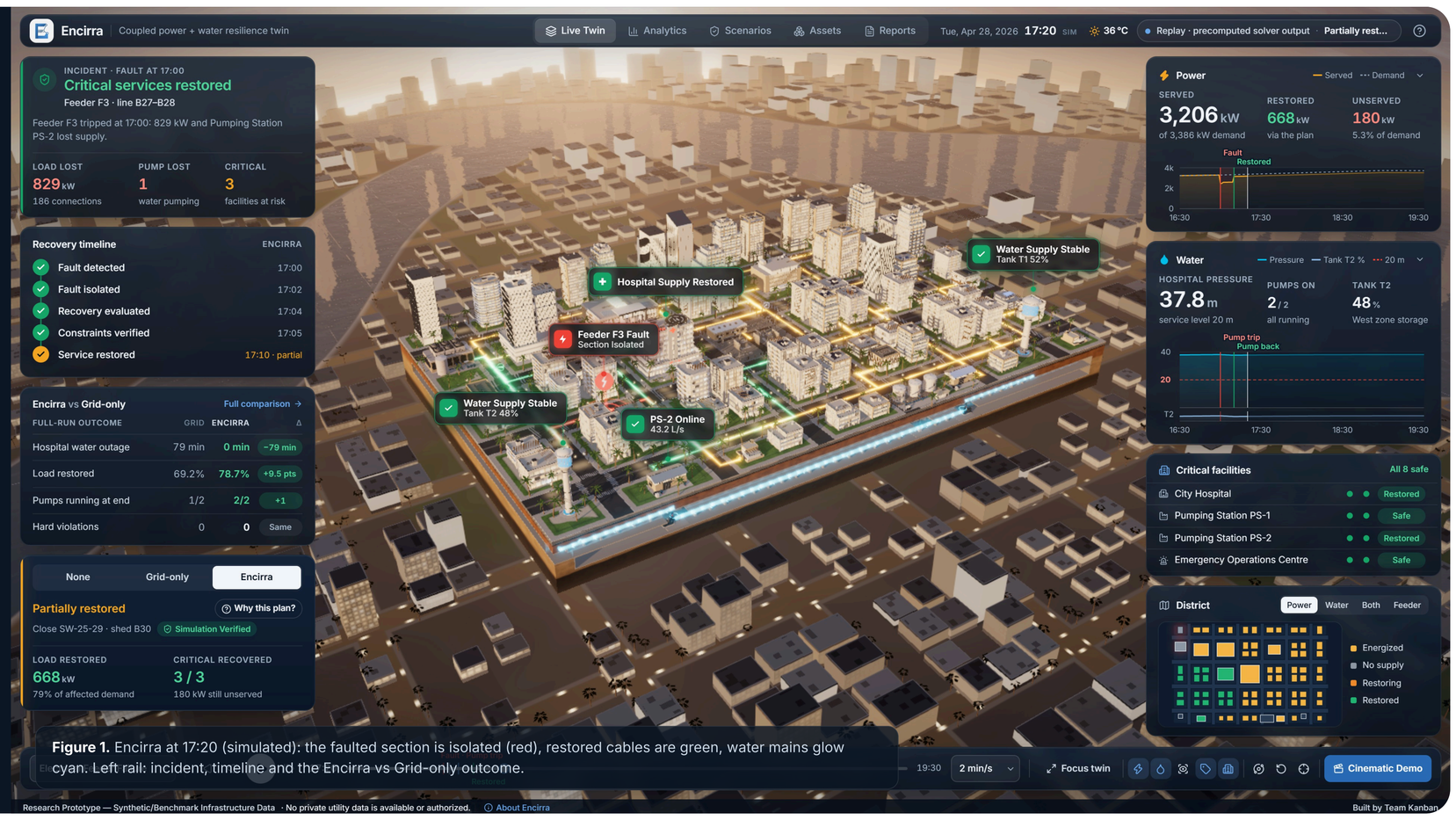


Figure 1. Encirra at 17:20 (simulated): the faulted section is isolated (red), restored cables are green, water mains glow cyan. Left rail: incident, timeline and the Encirra vs Grid-only outcome.

01 The engineering gap

Restoring electrical load is not the same as restoring essential urban services.

Water pumps are electrical loads. When a feeder fails, elevated storage keeps pressure up for a while, then critical facilities lose supply. Power-only restoration counts a hospital on its genset as "served" and treats a pump as an ordinary load, so it can choose a plan that is electrically optimal and still leaves the taps dry.

FEEDER FAULT supply lost → **PUMP TRIPS** power → flow → **HEAD FALLS** tank drains

02 Coupled infrastructure model

ELECTRICAL
IEEE 33-bus feeder (Baran–Wu), 37 switchable lines, 5 ties · AC power flow

COUPLING
Pump runs only if its bus is energised and V ≥ 0.90 pu · power fed back

HYDRAULIC
2 zones, 2 pumps, 2 tanks · pressure-driven EPS, 60 s step

$$P_{pump}\,[kW] = \rho\, g\, Q\, H\, /\, (1000\, \eta)$$

Q [m³/s] · H [m] · η [–] · ρ [kg/m³] — iterated with the power flow to a fixed point

03 Evaluation design

Identical physics and inputs for all policies: fault at 17:00, 1-min frames over a 3-hour horizon, deterministic seed.

No action	The fault persists; electrical and hydraulic consequences are tracked.
Grid-only	Maximise restored electrical energy within hard limits (conventional restoration).
Encirra	Rank plans on the coupled power–water outcome; verify before applying.

04 Computational workflow

1. **Fault scenario** `scenario JSON` — Feeder event, load and demand profiles
2. **AC power flow** `pandapower` — Bus voltages, line loading, energisation
3. **Pump dependency** `coupling` — Source bus → pump availability → P = ρgQH/η
4. **Hydraulic run** `WNTR · EPANET 2.2` — Pressures, tank levels, delivered flow
5. **Recovery search** `OR-Tools CP-SAT` — Radial switching + load-shed candidates
6. **Verify & report** `NetworkX + re-solve` — Independent rebuild, minute-by-minute checks

05 Optimisation & independent checks

$$\max J = w_L E_{served} + w_C E_{crit} + w_W V_{water} + w_H T_{crit,water} + w_S S_{tank} - w_{sw} N_{sw}$$

1.0	2.0	2.0	1.5	0.01	0.05
load /MWh	critical /MWh	water /1000 m³	crit. water /h	storage /%	switching /op

HARD — REJECT
Power-flow convergence
0.90 ≤ V ≤ 1.05 pu
Line loading ≤ 100 %
Radial & connected, fault isolated
Remote switching only

SCORED
Critical head ≥ 20 m
Tank storage margin
Water demand delivered
Critical electrical service
Switching effort

Weights are visible and configurable; hard limits are never traded against the score. Plans passing every hard check are labelled **Simulation Verified**.

06 Results · feeder-fault scenario

INDICATOR	NONE	GRID-ONLY	ENCIRRA
Load restored $R_L = 100 \cdot P_{rest}/(P_{rest}+P_{unserved})$	0	69.2	**78.7** %
Water demand served $WDS = 100 \cdot \Sigma q_{del}/\Sigma q_{req}$	78.4	78.4	**100** %
Critical facilities recovered	0/3	0/3	**3/3**
Min. critical pressure head $h_{min} = \min_{t,j \in crit} h_j(t)$	0.0	0.0	**37.8** m
Time below 20 m (hospital)	79	79	**0** min
Critical energy unserved	315	315	**18** kWh
Hard-constraint violations	0	0	**0**
Switch operations	0	4	**4**
Plans evaluated / rejected	—	80 / 59 · planned in ≈ 18 s	

Pump power loss: Encirra back-feeds PS-2 via tie 18–33 (98 kW shed) — hospital 0 min without water vs 79 min. **High demand:** both policies converge on the same plan (78.6 %). Encirra's advantage depends on tie capacity (sensitivity analysis in the repository).

HOSPITAL WATER SERVICE · 180-MIN HORIZON

None	101
Grid-only	101
Encirra	180



Figure 2. Grid-only at 18:20. Electrically verified, yet PS-2 stays dark: Tank T2 is empty and hospital pressure is 0 m, 72 min after the pump trip.



Figure 3. Underground view. Surface tiles lift away to expose the mains, pumps and valves with live flow and pressure state.

07 Status & contribution

IMPLEMENTED
Coupled solvers, recovery planner, independent verification, interactive 3D twin (web)

TESTED
IEEE 33-bus benchmark reproduced (V_{min} 0.913 pu, 202.7 kW losses); 90 + 57 + 20 automated tests

LIMITS
Synthetic water network & ratings; balanced loads; switching + shedding actions only

NEXT
Valve actions, larger real topologies, demand & repair uncertainty, utility validation

References — [1] M. E. Baran, F. F. Wu, IEEE Trans. Power Delivery, 1989 · [2] L. Thurner et al., "pandapower," IEEE Trans. Power Systems, 2018 · [3] K. A. Klise et al., WNTR, Environ. Model. Softw., 2017 · [4] L. Rossman et al., EPANET 2.2, U.S. EPA, 2020 · [5] Google OR-Tools (CP-SAT) · [6] NetworkX.
Research prototype on benchmark and synthetic data; "Simulation Verified" is not a field certification.

Built by Team Kanban

Watch the demo
youtu.be/CouV6JQUuKk

Try it live
encirra.kanbanstudios.ae